Latex Brunson Poster

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Problem 2

We are trying to compute the 2-modulus of the family Γ, which represent the collection of walks or routes in a network. The purpose of this modulus is to see how easily a family can trasmit a signal through the network. To calculate the 2-modulus, we assign a density $\rho(e)$ to each edge e in the network. The goal then becomes to Since all the edge weights are equal, the optimization problem becomes

$$\begin{aligned} \text{minimize} \quad & \rho^T \rho \\ \text{subject to} \quad & N\rho \geq \mathbf{1} \end{aligned}$$

Solution

The objective function becomes

$$f(\rho) = \rho^T \rho = {\rho_1}^2 + {\rho_2}^2 + {\rho_3}^2 + {\rho_4}^2.$$

This represents the total squared density assigned to the four edges. From this, we can find the desnity ρ that makes this as small as possible. We start with the path matrix:

$$N = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

Thus,

$$N\rho = \begin{bmatrix} \rho_1 + \rho_2 \\ \rho_1 + \rho_3 + \rho_4 \end{bmatrix}.$$

So the constraint becomes

$$\begin{cases} \rho + 1 + \rho_2 \geq 1, \\ \rho_1 + \rho_3 + \rho_4 \geq 1. \end{cases}$$

The first constraint corresponds to the path $\gamma_1 = acd$ and the second constraint corresponds to the path $\gamma_2 = acdb$. It is important to note that each path must have a total density of 1.

Critical Points

We must first check whether the minimum for ρ can occur at an interior critical point, conducting multivariable calculus. For differentiable functions, the interior must satisfy:

$$\nabla f(\rho) = 0$$

This makes the gradient of the function

$$\nabla f(\rho) = \begin{bmatrix} 2\rho_1 \\ 2\rho_2 \\ 2\rho_3 \\ 2\rho_4 \end{bmatrix}.$$

Setting,

$$\nabla f(\rho) = 0.$$

yields

$$2\rho_1 = 2\rho_2 = 2\rho_3 = 2\rho_4 = 0$$

and so

$$\boxed{\rho = (0, 0, 0, 0)}.$$

However, $\boxed{(}\, 0, 0, 0, 0)$is not feasible since $0 + 0 = 0 < 1$

Thus, the constrained minimum cannot occur at the interior critical point and we must look at the boundaries instead.

Boundary 1 Point

At boundary 1, we see that the first path constraint comes into play. The inequality becomes the equality:

$$\rho_1 + \rho_2 = 1 \quad \Longrightarrow \quad \rho_2 = 1 - \rho_1.$$

After substituting into the objective function,

$$f_1(\rho_1) = \rho_1^2 + (1 - \rho_1)^2$$

Taking the derivative,

$$\frac{\partial f_1}{\partial \rho_1}(\rho_1) = 2\rho_1 - 2(1 - \rho_1) = 4\rho_1 - 2$$

After setting it equal to zero,

$$4\rho_1 - 2 = 0$$

which gives

$$\rho_1 = \frac{1}{2}, \qquad \rho_2 = \frac{1}{2}.$$

From this, we find the critical point from the first boundary to be

$$\boxed{\left(\frac{1}{2}, \frac{1}{2}, \rho_3, \rho_4\right)}.$$

This boundary provides us with a critical point, but only for the first two coordinates. We must look at the second path constraint for other candidates in this problem.

Boundary 2

The second boundary occurs where the second path constraint comes into play.

$$\rho_1 + \rho_3 + \rho_4 = 1$$

For a fixed value of ρ_1,

$$\rho_3 + \rho_4 = 1 - \rho_1.$$

The minimum of

$$\rho_3^2 + \rho_4^2$$

occurs when

$$\rho_3 = \rho_4 = \frac{1 - \rho_1}{2}.$$

The sum of the squared values are now minimized.Now, we can express ρ_3 and ρ_4 in terms of ρ_1, turning this into a one-variable optimization problem. The objective function becomes

$$f_2(\rho_1) = \rho_1^2 + 2\left(\frac{1 - \rho_1}{2}\right)^2.$$

Simplifying,

$$f_2(\rho_1) = \rho_1^2 + \frac{(1 - \rho_1)^2}{2}.$$

Taking the derivative,

$$\frac{\partial f_1}{\partial \rho_1}(\rho_1) = 2\rho_1 - (1 - \rho_1) = 3\rho_1 - 1$$

Setting the derivative equal to zero,

$$3\rho_1 - 1 = 0$$

gives

$$\rho_1 = \frac{1}{3}$$

So the corresponding point is

$$\rho_3 = \rho_4 = \frac{1 - \frac{1}{3}}{2} = \frac{1}{3}$$

From this, the critical point on the second boundary is

$$\boxed{\left(\frac{1}{3}, \rho_2, \frac{1}{3}, \frac{1}{3}\right)}.$$

Like before, we must check the final path constraint for all candidates in our original problem.

Intersection Between the Boundaries

This is the final boundary, where both constraints are active simultaneously. The inequalities would hold with the equality:

$$\begin{cases} \rho_1 + \rho_2 = 1, \\ \rho_1 + \rho_3 + \rho_4 = 1 \end{cases}$$

From the first constraint,

$$\rho_2 = 1 = \rho_1.$$

From the second constraint,

$$\rho_2 = 1 - \rho_1.$$

Like mentioned before, to minimize the quanityt $\rho_3^2 + \rho_4^2$ over a fixed value of ρ_1, ρ_3 and ρ_4 must be equal. So,

$$\rho_3 = \rho_4 = \frac{1 - \rho_1}{2}.$$

Through all of this, we can write every coordinate of ρ through the single variable ρ_1. After substituting, we get an objective function of

$$f(\rho_1) = \rho_1^2 + (1 - \rho_1)^2 + 2\left(\frac{1 - \rho_1}{2}\right)^2.$$

Simplifying,

$$f(\rho_1) = \rho_1^2 + \frac{3}{2}(1 - \rho_2)^2.$$

The Boundary Intersection Critical Point can be found through

$$\frac{\partial f_1}{\partial \rho_1}(\rho_1) = 2\rho_1 + 3(1-\rho_2) = 5\rho_1 - 3.$$

Setting

$$\frac{\partial f_1}{\partial \rho_1}(\rho_1) = 0$$

gives

$$5\rho_1 - 3 = 0$$

and therefore,

$$\rho_1 = \frac{3}{5}.$$

The other coordinates are found through

$$\rho_2 = 1 - \frac{3}{5} = \frac{2}{5},$$

and

$$\rho_3 = \rho_4 = \frac{1 - \frac{3}{5}}{2} = \frac{1}{5}$$

Therefore,

$$\boxed{\rho^* = \begin{bmatrix} \frac{3}{5} \\ \frac{2}{5} \\ \frac{1}{5} \\ \frac{1}{5} \end{bmatrix}}.$$

Calculate the Minimum/Candidates Test

We now evaluate the original objective function at ρ^*

$$f(\rho^*) = \left(\frac{3}{5}\right)^2 + \left(\frac{2}{5}\right)^2 + \left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^2.$$

$$= \frac{9}{25} + \frac{4}{25} + \frac{1}{25} + \frac{1}{25} = \frac{15}{25} = \frac{3}{5}.$$

Thus,

$$\boxed{\text{Mod}_2(\Gamma) = \frac{3}{5}}.$$

Problem 3

In this problem, we are attempting to maximize the dual function

$$g(\lambda) = \lambda^T \mathbf{1} - \frac{1}{4}\lambda^T N N^T \lambda.$$

The path matrix is

$$N = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}.$$

Therefore,

$$NN^T = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}.$$

Expanding the function, we get

$$g(\lambda_1, \lambda_2) = \lambda_1 + \lambda_2 - \frac{1}{2}\lambda_1^2 - \frac{1}{2}\lambda_1\lambda_2 - \frac{3}{4}\lambda_2^2.$$

Critical Points

The first thing we must do is get the critical points in the interior of the feasible region, We can do this by setting both partial derivatives equal to zero.

$$\frac{\partial g}{\partial \lambda_1} = 1 - \lambda_1 - \frac{1}{2}\lambda_2$$

$$\frac{\partial g}{\partial \lambda_2} = 1 - \frac{1}{2}\lambda_1 - \frac{3}{2}\lambda_2.$$

Setting both partial derivatives equal to zero,

$$\begin{cases} 1 - \lambda_1 - \frac{1}{2}\lambda_2 = 0, \\ 1 - \frac{1}{2}\lambda_1 - \frac{3}{2}\lambda_2 = 0 \end{cases}$$

Solving this system gives,

$$\boxed{\lambda_1 = \frac{4}{5}, \qquad \lambda_2 = \frac{2}{5}}$$

This critical point is feasible because both values are nonnegative

Boundary Points

We must also check the boundaries of the feasible region.

$$\text{The boundaries are} \lambda_1 = 0 \text{and} \lambda_2 = 0 \text{because} \lambda_1, \lambda_2 \geq 0$$

Boundary 1:

For

$$\lambda_1 = 0$$

$$g(0, \lambda_2) = \lambda_2 - \frac{3}{4}\lambda_2^2$$

We can then find the critical point along this boundary by taking the derivative

$$\frac{d}{d\lambda_2} g(0, \lambda_2) = 1 - \frac{3}{2}\lambda_2.$$

Thus,

$$\lambda_2 = \frac{2}{3}$$

Thus, the candidate point on this boundary is

$$\left(0, \frac{2}{3}\right).$$

Boundary 2:

For

$$\lambda_2 = 0$$

$$g(\lambda_1, 0) = \lambda_1 - \frac{1}{2}\lambda_1^2.$$

$$\frac{d}{d\lambda_1} g(\lambda_1, 0) = 1 - \lambda_1$$

Thus,

$$\lambda_1 = 1$$

Thus, the candidate point on this boundary is

$$(1, 0)$$

Candidates Test

We can now evaluate g at the interior critical point and the two boundary candidates.

For the interior critical point,

$$g\left(\frac{4}{5},\frac{2}{5}\right) = \frac{4}{5}+\frac{2}{5}-\frac{1}{2}\left(\frac{4}{5}\right)^2-\frac{1}{2}\left(\frac{4}{5}\right)\left(\frac{2}{5}\right)-\frac{3}{4}\left(\frac{2}{5}\right)^2$$

$$= \boxed{\frac{3}{5}}.$$

For the boundaries

$$g\left(0,\frac{2}{3}\right) = \frac{2}{3}-\frac{3}{4}\left(\frac{2}{3}\right)^2 = \frac{1}{3}.$$

$$g(1,0) = 1-\frac{1}{2} = \frac{1}{2}.$$

Therefore,

$$\frac{3}{5} > \frac{1}{2} > \frac{1}{3},$$

so the maximum is

$$\boxed{\lambda^* = \left(\frac{4}{5},\frac{2}{5}\right)}$$

and the maximum value is

$$\boxed{\max_{\lambda\geq 0} g(\lambda) = \frac{3}{5}.}$$

Problem 4

$$\mathrm{Mod}_{2,\sigma}(\Gamma) = \min_{\rho}\sum_{e\in E}\sigma(e)\rho(e)^2$$

subject to

$$N\rho \geq \mathbf{1}.$$

Simply put, we are attempting to find the density ρ that makes the sum of squared densities as small as possible. Specifically, $\sigma(e) = 1$ for every edge, so the formula becomes

$$\mathrm{Mod}_2(\Gamma) = \min_{\rho}\left(\rho_1^2+\rho_2^2+\rho_3^2+\rho_4^2\right)$$

subject to

$$N\rho = \begin{bmatrix} \rho_1 + \rho_2 \\ \rho_1 + \rho_3 + \rho_4 \end{bmatrix} \geq \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

At the intersection where both constraints are equalities, as shown in Problem 2, we have

$$\rho_1 + \rho_2 1$$

and

$$\rho_1 + \rho_3 + \rho_4 = 1$$

From this,

$$\rho_2 = 1 - \rho_1$$

and

$$\rho_3 + \rho_4 = \rho_1$$

When solving for these equations, we find that the minimum of $\rho_3^2 + \rho_4^2$ occurs when

$$\rho_3 = \rho_4 = \frac{1 - \rho_1}{2}$$

In other words, through this equation, we are able to turn this four-variable problem into a one-variable problem. When simplifying, we get,

$$f(\rho_1) = \rho_1^2 + \frac{3}{2}(1 - \rho_1)^2.$$

We next take the derivative of this equation so we can find our specific value at the zero;

$$f'(\rho_1) = 2\rho_1 - 3(1 - \rho_1) = 5\rho_1 - 3$$

When setting the derivative equal to zero, we get

$$5\rho_1 - 3 = 0$$

so

$$\rho_1 = \frac{3}{5}$$

and

$$\rho_2 = 1 - \frac{3}{5} = \frac{2}{5}$$

Now, all we have to do is plug the values back into the intiial modulus to find $\mathrm{Mod}_2(\Gamma)$. Specifically, the extremal density should be.

$$\boxed{\rho^* = \left(\frac{3}{5}, \frac{2}{5}, \frac{1}{5}, \frac{1}{5}\right)}.$$

The 2-modulus is the value of the objective function at ρ^*:

$$\text{Mod}_2(\Gamma) = \left(\frac{3}{5}\right)^2 + \left(\frac{2}{5}\right)^2 + \left(\frac{1}{5}\right)^2 + \left(\frac{1}{5}\right)^2$$

Therefore,

$$\text{Mod}_2(\Gamma) = \frac{9}{25} + \frac{4}{25} + \frac{1}{25} + \frac{1}{25} = \frac{15}{25} = \boxed{\frac{3}{5}}.$$

This confirms that the final result of $\text{Mod}_2(\Gamma)$ is $\frac{3}{5}$

Problem 5

For this problem, we have $p = 2$, meaning that the graph has two paths. Path 1 is $\gamma_1 = acb$ and path 2 is $\gamma_2 = acdb$. From this, we know that there are two Lagrange multipliers, where $\lambda_1 = \lambda_(\gamma_1)$ and $\lambda_2 = \lambda_(\gamma_2)$. Considering that all edge weights are 1, the Lagrangian is

$$L(\rho, \lambda) = \rho_1^2 + \rho_2^2 + \rho_3^2 + \rho_4^2 + \lambda_1(1 - \rho_1 - \rho_2) + \lambda_2(1 - \rho_1 - \rho_3 - \rho_4)$$

FOr a fixed λ, we minimize L with respect to ρ. This leads us to the important part of stationarity, in which we take the derivative with respect to each ρ_i, set the derivatives equal to zero, and solve. Thus, we can find the ρ that minimizes the Lagrangian.

Again, in repeating the conditions, we have

$$\frac{\partial L}{\partial \rho_1} = 2\rho_1 - \lambda_1 - \lambda_2 = 0$$

$$\frac{\partial L}{\partial \rho_2} = 2\rho_2 - \lambda_1 = 0$$

$$\frac{\partial L}{\partial \rho_3} = 2\rho_3 - \lambda_2 = 0$$

$$\frac{\partial L}{\partial \rho_4} = 2\rho_4 - \lambda_2 = 0$$

When solving these equations to isolate individual ρ_i, we get,

$$\rho_1 = \frac{\lambda_1 + \lambda_2}{2}, \qquad \rho_2 = \frac{\lambda_1}{2}, \qquad \rho_3 = \frac{\lambda_2}{2} \qquad \rho_4 = \frac{\lambda_2}{2}$$

$$\boxed{\rho_\lambda = \left(\frac{\lambda_1 + \lambda_2}{2}, \frac{\lambda_1}{2}, \frac{\lambda_2}{2}, \frac{\lambda_2}{2}\right)}.$$

After reaching this stationarity condition, where $\rho_(\lambda)$ minimizes the Lagrangian for a given λ, we must plug it back in.

$$g(\lambda) = \lambda_1 + \lambda_2 - \frac{1}{4}\left(2\lambda_1^2 + 2\lambda_1\lambda_2 + 3\lambda_2^2\right).$$

Therefore,

$$g(\lambda) = \lambda_1 + \lambda_2 - \frac{1}{2}\lambda_1^2 - \frac{1}{2}\lambda_1\lambda_2 - \frac{3}{4}\lambda_2^2$$

From there, ρ is eliminated, and we are left with λ_1 and λ_2. We must maximimize $g(\lambda)$ over $\lambda_1, \lambda_2 \geq 0$, as the problem states. With the stationairty conditions as

$$\frac{\partial g}{\partial \lambda_1} = 1 - \lambda_1 = \frac{1}{2}\lambda_2 = 0$$

$$\frac{\partial g}{\partial \lambda_2} = 1 - \frac{1}{2}\lambda_! - \frac{3}{2}\lambda_2 = 0$$

Solving this gives us

$$\lambda_1^* = \frac{4}{5} \qquad \lambda_2^* = \frac{2}{5}.$$

Thus,

$$\lambda^* = \left(\frac{4}{5}, \frac{2}{5}\right).$$

When completing the dual objective at λ^*, we get the final calculation that,

$$\mathrm{Mod}_2(\Gamma) = g(\lambda^*)$$

$$= \frac{4}{5} + \frac{2}{5} - \frac{1}{2}\left(\frac{4}{5}\right)^2 - \frac{1}{2}\left(\frac{4}{5}\right)\left(\frac{2}{5}\right) - \frac{3}{4}\left(\frac{2}{5}\right)^2$$

Therefore,

$$\mathrm{Mod}_2(\Gamma) = \frac{6}{5} - \frac{8}{25} - \frac{4}{25} - \frac{3}{25} = \frac{15}{25} = \boxed{\frac{3}{5}}.$$

Therefore, both methods, the direct simplification method and the Lagrangian method, yield the same results. This is as expected since all the Lagrangian method does is replace the four unknown ρ_i with two λ_i to simplify the problem.